

Khushi Gupta

Fourth Year Undergraduate

Department of Computer Science and Engineering | Indian Institute of Technology Kanpur

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2022-Present	B.Tech	Indian Institute of Technology Kanpur	9.1/10
2021-22	Class XII (CBSE)	St. Xaviers School, Delhi	96%
2019-20	Class X (CBSE)	St. Xaviers School, Delhi	95.6%

Achievements & Scholarships

- Received **Academic Excellence Award** for exceptional academic performance from IIT Kanpur in 2022-23 and 2024-25
- Achieved **All India Rank of 751** in common rank list in **JEE Advanced 2022** among 2.5 Lakh shortlisted candidates
- Achieved **All India Rank of 815** with **100 percentile** in Physics in **JEE Mains 2022** among 1 million+ candidates
- Received **Certificate of Merit** from CBSE for outstanding performance in **Mathematics** and **Physics** in AISSCE 2022
- Ranked **10th** out of **100+** teams from all over India in **HackIITK CTF** organised by C3i Hub and SnT Council IITK

Competitive Programming

- Achieved a **Global Rank 42** in **ICPC Algo Queen 2024** among **237** out of **4500** females shortlisted in the first round
- Achieved a maximum rating of **1801 (Expert)** on **Codeforces** (Profile - Shi_07g)
- Achieved a maximum rating of **2022** on **CodeChef**, **Global Rank 38** in CodeChef Starters 181(Div 2)(Profile-khushi07g)
- Achieved **Global Rank 318** in Codeforces round 954 and **343** in Codeforces round 1019 among **16000** official participants
- Ranked **72nd** out of **17000** participants in the **IICPC Codefest' 24** Prelims, a pan-India individual coding contest

Work Experience

Software Engineering Intern | Graviton Research Capital, India

(May'25 - Jul'25)

- Designed and implemented a **UDP simulation socket and parser** to replay captured network traffic from PCAP files
- Added support for both **unicast** and **multicast UDP connections** in the simulation socket parser
- Automated simulation setup using **Python scripts**, including syncing necessary files, logs, and generating configuration files
- Designed and built a **Tshark replacement tool** using the **PcapPlusPlus** library, for parsing large PCAP files efficiently
- Focused on efficient **TLS decryption** and **WebSocket stream parsing** to produce structured and human-readable output
- Achieved about **3× speedup over Tshark** by minimizing overhead and optimizing for fast, focused packet processing

STEP Intern | Google Cloud Databases | Google, India

(May'24 - Jul'24)

- Developed a **Python-based tool** to detect and extract **stack traces** from server logs for early detection of customer issues
- Integrated **shell scripts** and **gcloud CLI** to automate trace extraction and generate **real-time alerts** for on-call engineers
- Analyzed **stack trace patterns** and created detailed **design documents** and **case studies** for streamlined debugging
- Helped identify critical issues like **segmentation faults** & **server health problems**, significantly reducing debugging time

Key Projects

Building GemOS | Course Project: Operating Systems | Prof. Debadatta Mishra

(Aug'24 - Nov'24)

- Implemented a user-space allocator with **malloc()/free()** built on **mmap()** and linked-list management
- Added exception-handling features: **instruction skip**, **function skip**, and **gaining OS privilege**
- Developed user-level utilities to inspect memory segments, view virtual memory mappings, and implemented **trace buffer**
- Developed a **round-robin scheduler** to manage processes and implemented an iterative **page table walker** for translation
- Integrated the **mremap()** syscall by allocating new pages, updating page tables, and flushing the **TLB**
- Provided full semaphore operations using **sem_wait()** and **sem_post()** to coordinate thread synchronization

Parallel Local Extrema Detection | Course Project: Parallel Computing | Prof. Preeti Malakar 🌐

(Jan'25 - Apr'25)

- Designed **3D domain-decomposed MPI** framework for time-series volume data featuring optimized local extrema detection
- Implemented **parallel I/O** with an **odd-even communication** pattern to streamline halo exchanges among processes
- Deployed on the **PARAM Rudra cluster** under **SLURM**, evaluating scalability across varying data sizes & process counts

Programming for Performance and Analysis | Self Project | 🌐

(July'25 - Sep'25)

- Built a C++ producer-consumer pattern using synchronization primitives like **std::unique_lock** & **std::condition_variable**
- Diagnosed cache contention & bottlenecks using profilers like **PAPI** & **perf c2c**, resolving them by cache-aware programming
- Implemented a **two-level cache simulator** in C++ with **LRU**, **quad-age** and **Belady's optimal** replacement policies
- Analyzed miss rate variations in **L2/L3** caches for **inclusive**, **exclusive**, **NINE**, and **hybrid** cache management protocols
- Designed **PC and address based multi-stream stride prefetchers** to improve prediction accuracy and cache hit rates

OptiQuery | Course Project: Principles of Database Systems | Prof. Arnab Bhattacharya 🌐

(Jan'25 - Apr'25)

- Designed a parser that translates **SQL queries** into **relational algebra trees** for systematic rule-based optimization
- Implemented **predicate pushdown** and **join reordering** to substantially reduce estimated cost of complex query executions
- Developed an interactive UI that visualizes query execution plans and comparative cost metrics before and after optimization

Computer Networks | Course Project: Computer Networks | Prof. Adithya Vadapalli 🌐

(Jan'25 - Apr'25)

- Developed a **TCP-based** chat server that facilitates both group broadcasts and one-to-one messaging across multiple clients
- Implemented a custom **DNS resolver** capable of performing **iterative** and **recursive** lookups
- Implemented key network-layer routing protocols like **Distance Vector Routing (DVR)** and **Link State Routing (LSR)**

Random Walks & Electric Circuits | *Course Project: Randomized Algorithms* | Prof. Surender Baswana (Jan'25 - Apr'25)

- Analyzed connections between **random walks** and **electrical circuits**, relating commute times to effective resistance
- Proved via **linear algebra** that current configurations in resistor networks with external sources are unique
- Derived the final result $C_{uv} = 2mR_{uv}$ by applying the principles of **reversibility** and **superposition** of electric circuits

Pedal Pal | *Course Project: Software Development and Operations* | Prof. Indranil Saha (Jan'24 - Apr'24)

- Collaborated with a team of 10 students to develop a full-stack Android app for bicycle rentals inside the campus
- Utilized **Django** and **Django Rest Framework** for backend development and employed **Flutter** to create a responsive user interface, ensuring an optimal user experience. Efficiently handled and organized application data with **PostgreSQL**
- Systematically maintained the system requirement specification, design details, user manuals and testing documents

Companion Arbiter PUF | *Course Project: Intro to Machine Learning* | Prof. Purushottam Kar (Mar'24 - Apr'24)

- Collaborated with a team of 6 for the decryption of a **Physically Unclonable Function** (PUF) using linear classifiers by analyzing the difference in response time delay between ports against a threshold value in two chains of multiplexers
- Conducted in-depth analysis of training time and accuracy of two different models, **LinearSVC** and **Logistic Regression**

Verilog HDL Processor | *Course Project: Computer Organisation* | Prof. Mainak Chauduri (Mar'24 - Apr'24)

- Developed a **32-bit single cycle processor** using **Verilog**, simulated on **Xilinx ISE** and tested on Spartan 3E FPGA
- Designed instruction layout and encoding methodologies for R, I and J instructions in accordance with **32-bit MIPS ISA**
- Implemented a **seven-state FSM** for fetch-decode-execute cycle, working with program counter, ALU and memory unit

Quantum Lower Bounds | *Course Project: Complexity Measures for Boolean Functions* | Prof. Rajat Mittal (Aug'24 - Nov'24)

- Studied Quantum Adversary Method from **“Quantum Lower Bounds by Quantum Arguments”** by Andris Ambainis
- It gives **improved lower bounds** on quantum query complexity and introduces new tool for analyzing complexity measures
- Analyzed its application to problems like Unordered Search, Majority, and AND-OR functions as presented in the paper

Technical Skills

- **Programming:** C/C++, Python, Javascript
- **Machine Learning:** Numpy, Pandas, Matplotlib, Seaborn
- **Web Utilities and Other:** ReactJS, Django, Verilog, Git, Bash, Linux, SQL, $\LaTeX$, perf, gdb

Relevant Coursework (# - ongoing)

Data Structures and Algorithm	Fundamentals of Computing	Linear Algebra & ODE
Introduction to Machine Learning	Discrete Mathematics	Software Development & Operations
Design & Analysis of Algorithms	Computer Organisation	Probability for Computer Science
Computer Networks	Operating Systems	Complexity Measures for Boolean Functions
Randomized Algorithms	Parallel Computing	Principles of Database Systems
Programming for Performance#	Principles of Programming Languages#	Multi Core & Multiprocessor Architecture#

Positions of Responsibility

Tutor | *Data Structures and Algorithms* (Aug'25 - Present)

- Assisted in designing and evaluating **theoretical** and **programming assignments**, reinforcing key concepts for students
- Administered and supported course activities for **300+ students**, including quiz and exam invigilation and evaluation

Academic Mentor | *Fundamentals of Computing* | *Counselling Service* (Aug'23 - Apr'24)

- Served as a mentor to first-year students, focusing on **ESC111 and ESC112 courses - Fundamentals of Computing**
- Provided personalized mentoring, throughout the first two semesters and organized extra classes at **institutional level**

Secretary | *Fine Arts Club* | *IIT Kanpur* (Jul'23 - Apr'24)

- Organized a diverse range of activities and competitions and guided a group of 4 students for the Freshers' Showcase
- Facilitated workshops to foster creativity within the campus community, providing hands-on experience to the students